
IC 140: Engineering Graphics

Course Code	IC 140
Course Name	Engineering Graphics
L-T-P-C	2-0-3-4
Prerequisites	Consent of the faculty member
Intended for	B.Tech 1st year
Distribution	Core
Approval	2nd Adhoc Senate (May 2012), in the B.Tech Institute Core (common core courses) curriculum
Revised	47th BoA (Sep 2022) and 37th Senate (Oct 2022) - renamed from “Graphics for Design”

Course Objectives

To teach basic concepts of engineering drawing with emphasis on improving students' imagination.

Course Contents

- **General:** Introduction to design process and drawings, sheet layout, line symbols, line groups, preferred scales, reference planes and quadrants, technical sketching; dimensioning, tools of dimensioning, size and position dimensions. Freehand sketching and mechanical drafting. (2 Lectures; 1 Practical)
- Introduction to CAD software for the creation of 3D models and 2D engineering drawings. (Take home assignments will continue throughout the semester) (4 Lectures; 2 Practical)
- **Projections:** Types of projections, theory of orthographic projections, projection of points, lines; oblique planes. (Free hand / Solid Works) (2 Lectures; 1 Practical)
- Projection of plane figures. (MD/Solid Works) (2 Lectures; 1 Practical)
- Projection of solids and sections. (MD/Solid Works) (2 Lectures; 1 Practical)
- Development of solids (MD/Solid Works) (2 Lectures; 1 Practical)
- Intersection of surfaces (MD/Solid Works) (2 Lectures; 1 Practical)
- Sketching of orthographic views from pictorial views. (2 Lectures; 1 Practical)
- Missing Line, Missing View Exercises (Free hand/Solid Works) (2 Lectures; 1 Practical)
- **Pictorial Views:** Isometric and oblique views from multi-planar orthographic views. (Free hand/Solid Works) (2 Lectures; 1 Practical)
- Limits, fits and tolerances; Schematic and process flow diagrams; standard equipment and symbols. (2 Lectures; 1 Practical)
- Instrumentation and control diagrams; flow charts. (2 Lectures; 1 Practical)